TEXORA · SESSIONAL NOTES · STUDY SHEET

# Grading of Production Pattern and Marker Making

*Two bench skills in one study: grading the size-M master pattern of the basic shirt up and down the size range by the grade rules, and planning a marker for the graded set — with marker efficiency calculated exactly as in the TE-2207 theory sheet.*

TE-2208 · APPAREL MANUFACTURING TECHNOLOGY-1 SESSIONAL · STUDY 5 · LEVEL 2 TERM 2

## 1. Aim and place of the study

**Aim:** to grade the production pattern of the basic shirt (Study 3, size M) across its size range using the grade rules, verify the graded sizes against the specification table, and then plan a marker for a stated size mix and calculate its efficiency. Definitions and principles are in TE-2207 Topics 3 and 4; this sheet is the worked bench version, and every number in it agrees with those sheets.

## 2. Grading — the rules and the worked example

The grade rules are read off the specification table of the shirt (TE-2207 Topic 2 §1.1): the fixed step between consecutive sizes at each point of measurement.

| POINT OF MEASUREMENT | CHEST | BODY LENGTH | SHOULDER | SLEEVE LENGTH | COLLAR | HEM WIDTH |
|---|---|---|---|---|---|---|
| Grade per size step (cm) | 5 | 2 | 2 | 1.5 | 2 | 5 |

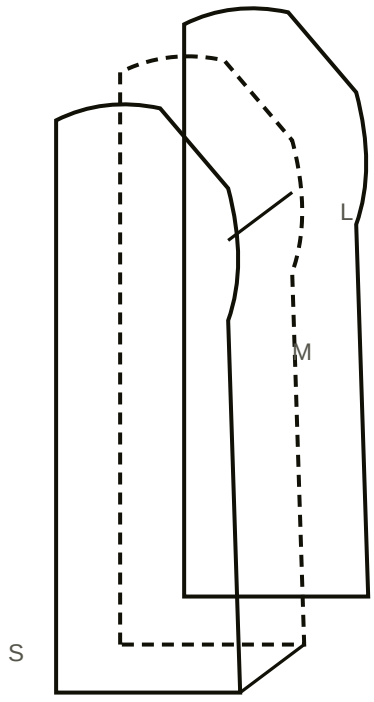
Measurement in size n = Base-size measurement + (number of steps × grade)

**Worked example.** The master pattern is size M. Grading one step up gives L; one step down gives S. Applying the rule at every point rebuilds the specification table — which is the verification that the grading is right:

| POINT (CM) | S (M − 1 STEP) | M (BASE) | L (M + 1 STEP) | XL (M + 2 STEPS) |
|---|---|---|---|---|
| Chest | 102 | 107 | 112 | 117 |
| Body length | 71 | 73 | 75 | 77 |

| | | | | |
|---|---|---|---|---|
| Shoulder width | 44 | 46 | 48 | 50 |
| Sleeve length | 60 | 61.5 | 63 | 64.5 |
| Collar size | 38 | 40 | 42 | 44 |
| Hem width | 100 | 105 | 110 | 115 |

**How the increments are applied at the pattern.** Each grade point of a piece moves by its share of the increment: the chest point moves out by one quarter of the chest grade (5 ÷ 4 = 1.25 cm per step on the half pattern), the shoulder point moves out by half the shoulder grade and up by its length share, the hem point moves down by the length grade. In manual (shift) grading the master piece is moved by these x–y amounts on a ruled table and the new outline is traced point by point; in CAD grading the same increments are stored as grade rules at the points. The check is always the same: rebuild the all-size measurement table from the graded patterns and compare it, cell by cell, with the specification.




The graded nest of the shirt front: each size is the master outline moved by the grade rules — out at the chest and shoulder points, down at the hem.

Separations are exaggerated so the increments can be seen; in the real nest the sizes lie almost on top of each other.

Figure 1 — Graded nest of the shirt front (S, M, L), drawn as in TE-2207 Topic 3 and produced at the bench in this study.

## 3. Marker making — the bench procedure

1. **Take the cutting plan:** sizes and quantities for this lay (for the exercise: 4 complete shirts — 2 × M, 1 × S, 1 × L — a stated size mix), the fabric's usable width (1.45 m for the exercise fabric), and any matching or nap rules (none for the exercise fabric — a plain weave, no nap).

2. **Assemble the pieces:** every piece of every garment in the mix, from the graded sets of §2 — completeness is checked against the Study 4 table before a single piece is placed.
3. **Arrange within the width:** working on a frame of the usable width, place the large body pieces first, then fit sleeves, yokes, collars, cuffs and pockets into the spaces between them; pieces may rotate 180° but their grain lines stay parallel to the marker's length.
4. **Measure and iterate:** the marker length is the distance from the start to the last piece; rearrange until no obvious gap remains, then record the length and compute the efficiency (§4).
5. **Draw and mark the marker:** trace every outline cleanly; write size and piece name inside each outline; the marker is then ready to lay on a spread (TE-2207 Topic 5) in the cutting studies.

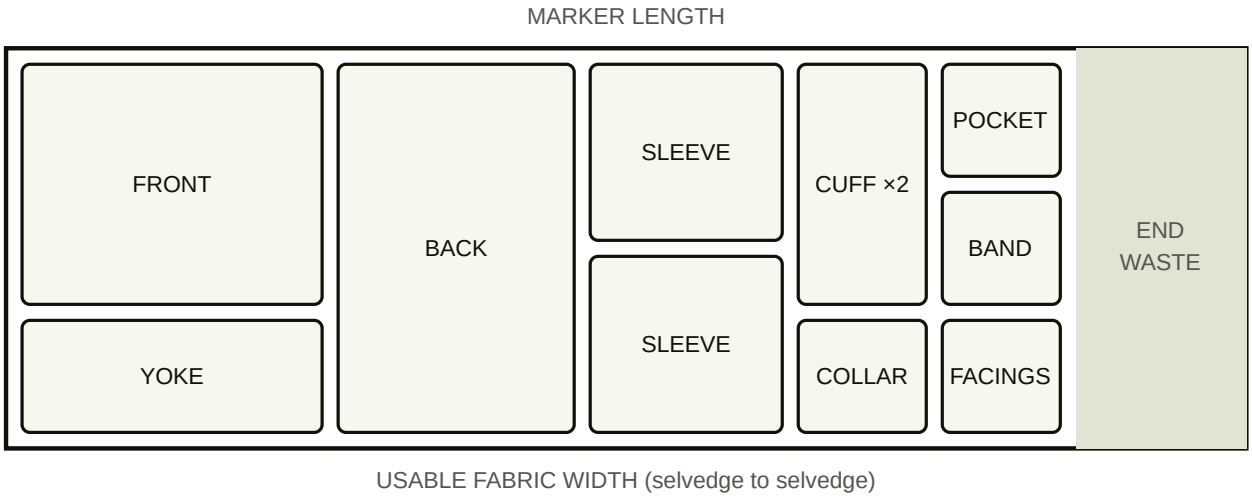



Figure 2 — Schematic marker for the shirt: pieces nested across the usable width; the uncovered areas are marker waste (the same plan as TE-2207 Topic 4, Figure 1).

## 4. Marker efficiency — the worked example of this course

$$\text{Marker efficiency (\%)} = (\text{Total area of pattern pieces} \div \text{Total area of the marker}) \times 100$$

$$\text{Marker area} = \text{Marker length} \times \text{Usable fabric width}$$

**Worked example (the course's standard example, as in TE-2207 Topic 4).** Usable width 1.45 m; marker length 6.5 m; the areas of all pattern pieces in the marker total 7.92 m².

- Marker area = 6.5 × 1.45 = 9.425 m²
- Marker efficiency = (7.92 ÷ 9.425) × 100 = **84.0%**
- Marker waste = 16.0% — about 1.51 m² in every marker length.
- The marker holds 4 complete garments, so fabric consumption = 6.5 ÷ 4 = **1.625 m per garment** of 1.45 m-wide fabric.

At the bench, piece areas are found from the CAD system where available, or estimated by dividing each piece into rectangles and triangles and summing — the method matters less than the discipline: every piece counted once, no piece counted twice, and the width used is the *usable* width between selvedges, never the nominal roll width.

## 5. Precautions

- Verify the graded set against the specification table before marker making — a marker faithfully multiplies a grading error into every garment of the lay.
- Keep grain lines parallel to the marker length on every piece, including the small ones fitted last into gaps; a pocket turned cross-grain to save 2 cm is a defect planned in advance.
- Count pieces in and count pieces drawn; the two counts must agree with the cutting plan before the marker leaves the table.
- Record marker length, width, size mix and efficiency on the marker itself — the cutting room and the costing office both work from these figures.

## 6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. From the grade rules, give the chest, length and collar of size XXL, taking M (107, 73, 40) as base. Show the steps.
2. A marker is 5.6 m long on fabric of usable width 1.50 m, and its pieces total 7.14 m². Calculate the efficiency and the waste percentage.
3. In the standard example of §4, how much fabric area is saved per marker length if rearrangement lifts efficiency from 84% to 88%?
4. Why is grading never done by enlarging the master pattern by a single percentage?
5. State two constraints that can force a marker's efficiency down even when the nesting is skilful.
6. A size is missing from a marker but the efficiency figure looks excellent. Explain the trap and the check that exposes it.
7. Why is the usable width, not the roll width, used in the marker area?

### Answer key

1. XXL is 3 steps above M. Chest = 107 + 3 × 5 = 122 cm; length = 73 + 3 × 2 = 79 cm; collar = 40 + 3 × 2 = 46 cm.
2. Marker area = 5.6 × 1.50 = 8.40 m². Efficiency = 7.14 ÷ 8.40 × 100 = 85.0%. Waste = 15.0%.
3. Saving = (0.88 − 0.84) × 9.425 ≈ 0.38 m² per marker length.

4. Because the body does not grow in proportion: chest steps 5 cm while length steps 2 cm and sleeve 1.5 cm. A uniform enlargement applies one ratio to all points and distorts every proportion the grade rules exist to protect.
5. Any two of: nap/pile direction (no 180° rotation freedom), check or stripe matching (pieces placed by the repeat), few large sizes in the mix (large pieces pack badly), narrow fabric width.
6. Efficiency measures coverage of the pieces that are present; a missing size raises coverage of the remainder and hides the omission. The check is the piece count against the cutting plan — completeness is verified before efficiency is quoted.
7. The selvedge strips cannot carry pattern pieces (they are denser, often flawed, and excluded from garments); area that can never hold a piece must not be counted as marker area, or the efficiency is understated and comparisons between fabrics are distorted.

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